

1. Feature Overview

The Listing Search feature allows users to search for newly built homes in specific cities, ZIP codes, or neighborhoods and refine the results using filters including price range and bedrooms. The system displays listings as cards and supports pagination or a “Load More” mechanism for browsing large datasets. The business objective is to deliver fast, accurate, and relevant listings that match user preferences.

2. Test Objectives:

- Validate basic search functionality for different types of location inputs.
- Confirm the correctness and boundaries of filters (price and bedrooms).
- Validate the interaction between filters and search, ensuring combined filtering narrows results appropriately.
- Confirm pagination or “Load More” preserves filter and search context across pages.
- Identify UX issues, incorrect data handling, or inconsistent states.
- Provide automation candidates suitable for CI/CD regression coverage.

3. In Scope

- Search for input validation (city).
- Data display within listing cards (price, bedrooms, location).
- Price range filter (min, max).
- Bedrooms filter
- Combined filters (price + bedrooms).
- Pagination / Load More behaviors.
- Result count and no results messaging.
- Cross-browser sanity (Chrome primary, one alternate browser).

4. Out of Scope

- User login, account management, and saved searches.
- Builder profile deep navigation beyond click-through.
- Filters are not part of the scope (bathrooms, amenities, etc.).
- Dedicated performance load testing.
- Accessibility audit beyond basic usability observations.
- Mobile device matrix beyond smoke validation.

5. Assumptions

- Public production environment is used; inventory may fluctuate.

- Data may not be deterministic, especially price and bedroom distribution.
- Pagination may use query parameters, infinite scroll, or hybrid UX.
- Filter state may be stored in UI state, URL parameters, or network requests.
- No test database is provided; SQL validation assumes reasonable schema.

6. Risks and Dependencies

- Risks:
 - Data volatility may cause false negatives in automated assertions.
 - UI layout (marketing-driven) may change frequently and break selectors.
 - Third-party dependencies (maps, analytics) may introduce intermittent noise.
 - Bot detection or throttling may interfere with test automation traffic.
- Mitigation Strategies:
 - Use robust automation locators (data attributes when available).
 - Apply partial validation rather than strict dataset equality checks.
 - Stub network responses for deterministic validation when applicable.
 - Monitor console/network for unexpected error signals.

7. Test Approach

- Manual Functional Testing:
 - Validate happy path search flows using high-inventory metro areas.
 - Verify boundary and negative filter behavior (min > max, zeros, extremes).
 - Verify user-facing no-results of messaging and behavior.
- Automation (Playwright):
 - Smoke: Basic location search return at least one valid listing card.
 - Filter: Price + bedrooms combined filtering with state validation.
 - Pagination: Verify state persistence across page transitions.
- Non-Functional:
 - Basic performance observation (page loads reasonably fast).
 - High-level usability clarity (labels, visibility, messaging).
 - Light accessibility checks (keyboard, labels, navigation).

8. Environment and Tools

- Environment: Production site
- Browsers: Chrome (latest stable)
- Tools: Chrome DevTools, Playwright, SQL client (assumed), screenshots

- Test Data Approach:
 1. Use popular metro areas (Dallas, Houston, Phoenix, Atlanta) to ensure sufficient results.
 2. Filter values such as 200k–400k or 300k–500k for price and 3+ for bedrooms.
 3. Document any stable data anchors discovered during exploration.

9. Acceptance Criteria

The feature must enable users to:

- Execute location-based search successfully.
- View listings that contain mandatory information (price, beds, location).
- Apply filters with consistent and predictable results.
- Navigate across pages without losing search context.
- Receive clear feedback in zero-result situations.

10. Prioritization

- P0 (Critical)
 - Basic search
 - Price filter correctness
 - Bedrooms filter correctness
 - Combined filtering
 - Pagination state handling
 - No-results handling
- P1 (High)
 - Invalid ranges (min > max)
 - Deep link behavior (if supported)
 - Browser back-button behavior
- P2 (Medium)
 - Extended usability and accessibility validation
 - Advanced filters not in initial scope